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PHYSICAL SCIENCES

Electronics with switchable flexibility for 3D conforming neural interfaces

Xingdao He^{1†}, Matthew Chamberlin^{1†}, Zhaoqi Chen^{2†}, Jing Li³, Yuyu Gao¹, Feng Guo¹, Yutao Ma¹, Xiaohu Wei⁴, Xuan Luo¹, Wencong Zhang⁵, Jin Qu¹, Dengshuai Wei⁶, Guangyu Zhu⁶, Feng Wang⁴, Han Zhang^{7,8}, Xinge Yu¹, Xi Chen³, Yong Yang^{2*}, Peng Shi^{1,9,10,11*}

The intricate cortical folds of large primates physically restrict access to substantial portions of neural information via interface devices. Here, we develop a bioelectronic system, sFlex-Fold, with switchable flexibility, representing the neural interface capable of nondestructive three-dimensional (3D) access to both cortical gyri and sulci, providing large-area, nonpenetrative deep tissue coverage. sFlex-Fold is based on an artificial intelligence (AI)-designed liquid metal alloy (LM-alloy), leveraging the phase change of the tailor-made LM-alloy to create neural interfacing electronics with tunable mechanical response to temperatures ranging from 25° to 37°C. The LM-alloy can be patterned into arbitrary circuit layouts with an ~10-micrometer resolution. The flexibility switching happens at the LM melting point, fine-tuned to 36.2°C, upon in vivo tissue contact, causing a three-order-of-magnitude reduction in the effective modulus of the implanted device. As a result, sFlex-Fold has the unique advantages of both a rigid and flexible state and can be morphed into complex, folded, 3D shapes. This enables nondestructive in vivo implantation into deep cortical sulci while maintaining large coverage (>80 square centimeters) over curved brain surfaces with tissue-matching mechanical compliance. Such 3D structural and mechanical mimicking enables high-quality electrical interfacing as quantitatively assessed using rodent and porcine models.

INTRODUCTION

Electronics with transient properties are redefining the function and life cycle of biomedical implants, offering safer, evolving, and adaptive solutions for many clinical needs (1, 2). This concept has profound impacts on the field of neural interfaces, which has seen remarkable advancements in recent years, driven by the potential to revolutionize the treatment of neurological disorders, enhance cognitive function, and enable direct communication between the brain and external devices (3–5). A successful neural interface must achieve high-fidelity recording and spatially precise stimulation of the nervous system. Penetrating electrodes, such as the Utah array (6) or Michigan probe (7), easily provide such electrical contact, but the use of this modality is fundamentally limited in a few key aspects. The rigid structure and invasive implantation of these electrodes provoke significant immune response and scar formation, which compromises their ability to achieve stable, long-term neural recording and stimulation (8–11). Their implantation is also physically destructive to native neural tissue, which inherently limits their capability to

study undisturbed neural circuits. In addition, most existing neural interfaces are effective for studying neural dynamics in small, localized regions of the brain, and their limited coverage restricts their ability to interplay with complex neural dynamics that cross multiple brain regions.

Large coverage flexible devices have been shown to effectively conform to protruding cortical gyri and the nonplanar surface of the brain (12–18); however, true conformation to whole cortical regions, including those hidden within deep sulcal folds, remains challenging as these devices do not have the rigidity to easily penetrate the folds. As two-thirds of the primate cortex resides within sulci (19, 20), devices unable to access these regions miss most cortical information (20), and complete coverage is essential for enhancing functionality and versatility (17). This is especially true for complex prosthetic tasks, requiring the integration of different types of neural information, such as motor control, sensory feedback, and cognitive processing (21).

Effective mechanical coupling between device and tissue, driven by intimate physical contact and modulus matching, is another key to success (9, 11). Rigid probes shift their position relative to the tissue during normal subject motion, which can disrupt the neural interface and cause unexpected tissue damage. The mechanical mismatch is also more likely to trigger chronic immune response, which reduces the functional life span of the device (22–24). Furthermore, neural interfaces should be capable of adaptive confirmation to the changing neuronal landscape and maintain electrical contact over time as alterations in cortical structure are a notable character of many neurodegenerative diseases, driven by pathological processes such as neuronal loss, synaptic degeneration, and protein accumulation (25, 26). These considerations emphasize the unmet need for neural interfaces capable of complete access to all information rich brain regions, including cortical gyri and sulci, by combining the merits of both rigid penetrating devices and large-coverage flexible electronics.

Toward these goals, we develop a bioelectronic system, sFlex-Fold, with switchable flexibility, that constitutes the neural interface

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capable of inherent, nondestructive three-dimensional (3D) neural access to both cortical gyri and sulci, providing large-area, nonpenetrating deep tissue coverage. sFlex-Fold is built on an artificial intelligence (AI)-designed liquid metal alloy (LM-alloy) and exhibits a tunable mechanical response across 25° to 37°C. This is achieved by leveraging the phase change of the tailor-made LM-alloy, allowing for the creation of neural interfacing electronic devices with high-resolution, arbitrarily patterned circuits. Its flexibility switching is provided by the phase transition between solid and liquid of the highly conductive LM-alloy core, which is derived by analyzing more than 30,000 alloy formulations to tune the melting point to the desired 37°C. Consequently, sFlex-Fold has the unique advantages of both rigid and flexible states and can be morphed into complex, folded, 3D shapes to enable nondestructive in vivo implantation into the deep cortical sulci while maintaining large-area coverage over curved brain surfaces with tissue-matched mechanical compliance. The 3D structural and mechanical mimicking enables high-quality electrical interfacing, which is quantitatively assessed in an ex vivo porcine brain model and demonstrated via high-fidelity in vivo neural recording and stimulation in a rodent model.

RESULTS

System design

The sFlex-Fold neural interface consists of both a mechanical backbone and functional electronic circuit, patterned out of an AI-designed LM-alloy and encapsulated by a thin polydimethylsiloxane (PDMS) film (Fig. 1). The unique phase change properties of sFlex-Fold allows for the device to switch from rigid-to-flexible (flexible-to-rigid) upon melting (solidification) of the LM-alloy core. The rigid state of sFlex-Fold not only benefits its microscale fabrication but also allows the sFlex-Fold to exist as a truly 3D electrode array, after being morphed into the shape of the desired cortical region. This provides sFlex-Fold with the ability to nondestructively access the deep cortical sulci regions, via shape memory and flexibility switching (movie S1), that are inaccessible to existing flexible neural interface devices. After brief contact with the tissue, the rigid-to-flexible switching provides it with the capabilities of mechanical matching and conformation to gyri. The contact warms the device above the optimized LM-alloy melting temperature (36.2°C) to enable structural, mechanical, and electrical 3D conformal contact with the cortex (Fig. 1).

Specifically, in this study, sFlex-Fold is molded onto a reconstructed 3D model of human brain. First, the rigid-to-flexible transition allows the device to become flexible and mold to the model. Once the desired cortex-mimicking folding is achieved, a brief cooling triggers flexible-to-rigid switching to preserve the optimized 3D shape and curvature. The folded device is then translated onto the subject's brain and is implanted as a rigid single component device, achieving high-quality electrical contact and placement. Upon the in vivo implantation, the body temperature-induced intrinsic warming of the device triggers a second rigid-to-flexible switching to render an extreme reduction in the effective modulus (2.03 GPa to 1.66 MPa) of sFlex-Fold, providing mechanical matching between device and tissue, maintaining nondestructive, large-coverage, and long-term neural interfacing (movie S2).

AI-powered alloy design for optimized phase transition

The switchable flexibility of the sFlex-Fold depends on the phase change of the LM-alloy. To optimize the performance of the system,

creation of an LM-alloy core with a near-body-temperature melting point was required. This was done by using an AI-powered eutectic alloy discovery system. Gallium (Ga) was chosen as the main component of the LM-alloy because of its low melting point and biocompatibility. An established machine learning architecture for alloy discovery was further trained with a dataset of known parameters of Ga and its alloys (fig. S1), which includes 173 compositions and corresponding melting peak temperatures (Fig. 2A). The machine learning algorithm used 17 data descriptors as the inputs (27) (table S1) and resulted in an artificial neural network (ANN) model capable of predicting the melting points of Ga-based alloys, such as Ga-Ag, Ga-Ti, Ga-Ti-Sn, Ga-Sn-In, Ga-In-Bi, etc. The ANN model was used to generate a library of 35,145 alloy formulations and their melting points (Fig. 2B). The generation of this library acts as a prescreening step, reducing experimental workload by removing the need for physical validation of likely noncontenders and producing a short list of predicted top performers. By predicting the thermal properties of over 30,000 formulations, this system allows for the rapid identification of alloys with specific melting points tailored to diverse use cases beyond body temperature. The model's held-out performance (fig. S1, B and C; root mean square error around $\pm 40^\circ\text{C}$, $R^2 \approx 0.98$) makes it extremely useful in shrinking the haystack of the vast empirical search space (0° to 1400°C melting landscape across tens of thousands of candidate compositions). After filtering for theoretical biocompatibility and conductivity, the top 3 performers matching the near-body-temperature melting point requirements were selected. Namely, $\text{Ga}_{97}\text{Ag}_3$, $\text{Ga}_{86}\text{Sn}_{14}$, and $\text{Ga}_{82}\text{In}_{10}\text{Zn}_8$ were selected for experimental validation (Fig. 2C). $\text{Ga}_{97}\text{Ag}_3$, $\text{Ga}_{86}\text{Sn}_{14}$, and $\text{Ga}_{82}\text{In}_{10}\text{Zn}_8$ alloys were therefore synthesized and experimentally characterized (Fig. 2D and fig. S2). On the basis of the melting point characterization using differential scanning calorimetry (DSC), the $\text{Ga}_{97}\text{Ag}_3$ alloy, hereafter termed GaAg, was chosen for fabricating the sFlex-Fold.

A well-alloyed, stable multiphase microstructure was confirmed by scanning electron microscopy with energy-dispersive x-ray spectroscopy (SEM-EDX). Two silver (Ag)-containing intermetallic phases, a particle-like phase and a needlelike phase are finely dispersed within a continuous Ga-rich matrix (fig. S3). Quantitative EDX also shows ~4.8% Ag in the particle-like phase and ~21.2% Ag in the needlelike phase, whereas the surrounding matrix contains ~0.2 to 0.4% Ag, consistent with a dissolved Ag fraction and absence of unalloyed Ag. The enhanced switching capability was enabled by increasing the melting point of the new LM-alloy from ~33.3° to ~36.2°C (melting peak temperature in comparison to Ga). The sharp peak of the melting point also ensures the ability of the device to have fast and efficient rigid-to-flexible switching precisely at the desired melting point (Fig. 2D). AI-powered alloy generation has also improved the flexibility switching of the device by reducing the supercooling effect present in the flexible-to-rigid switching (28). This can be seen through the DSC curve where the freezing point (onset temperature) has been increased from -32° to -21.8°C compared to Ga. Critically, tuning the melting temperature close to 37°C is necessary to ensure predictable mechanical switching during implantation, as further evidenced by simulated implantation dynamics. At a handling temperature of 32°C, GaAg devices retained structural rigidity during placement, whereas pure Ga, due to its lower melting point, underwent premature flexibility switching (fig. S4). The GaAg alloy also extended the working time before phase transition by ~2-fold compared to pure Ga (fig. S5), providing an increased temporal window for accurate device positioning before mechanical softening occurs.

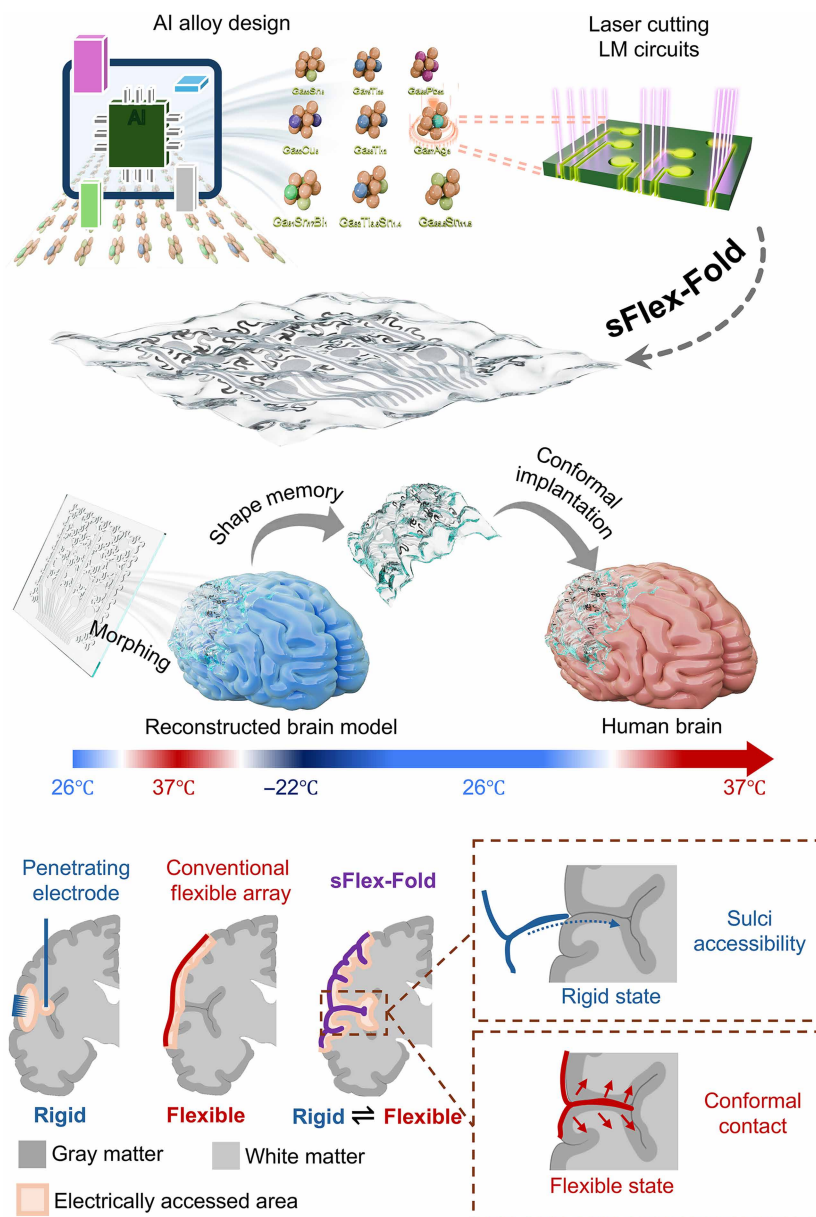


Fig. 1. Illustration of the sFlex-Fold. sFlex-Fold is capable of accessing deep cortical sulci as well as compliant, conformal contact in vivo, enabled by its unique switchable flexibility mechanism. The core of the device is an LM-alloy that was tuned through AI-guided alloy discovery to have a near-body-temperature melting point. The ability to pattern the LM-alloy with a low-energy laser, at a high resolution, allowing the device to be generalized to many applications with different circuit requirements. By using its temperature sensitive switchable flexibility, the planar sFlex-Fold can be easily handled at room temperature before morphing to the shape of a cortex model, the 3D shape of which is conserved by the device. Once transferred to the subject's brain, the rigid device inherently accesses the deep sulci, before undergoing rigid-to-flexible switching and maintaining conformal contact in its soft phase. Through this transitional flexibility, it establishes a paradigm of neural interfaces with both advantages of penetrating electrodes (rigidity for insertion) and flexible interfaces (soft, tissue-matched mechanics), which enables larger conformal coverage than existing strategies.

The LM-alloy core acts as both the electronic circuit and mechanical backbone of the sFlex-Fold; as a result, both the mechanical and electrical properties of GaAg were analyzed. The addition of highly conductive Ag to GaAg enhanced the conductivity of the alloy by 140% over Ga (Fig. 2E), verifying the capability of GaAg for high-quality neural recording and stimulation. In particular, for high-frequency applications (0.1 to 1 kHz), the impedance of a device made with GaAg is 37 times improved compared to that of Ga (Fig. 2F).

The mechanical testing revealed that there was no significant difference in the change of bending stress for devices made of GaAg or pure Ga (Fig. 2G).

Low-energy laser cutting for fabricating sFlex-Fold

We developed a method of patterning the GaAg, which is easily implemented, of high resolution, and generalizable to arbitrary circuit patterns (Fig. 3A). The fabrication process begins with spin-coating

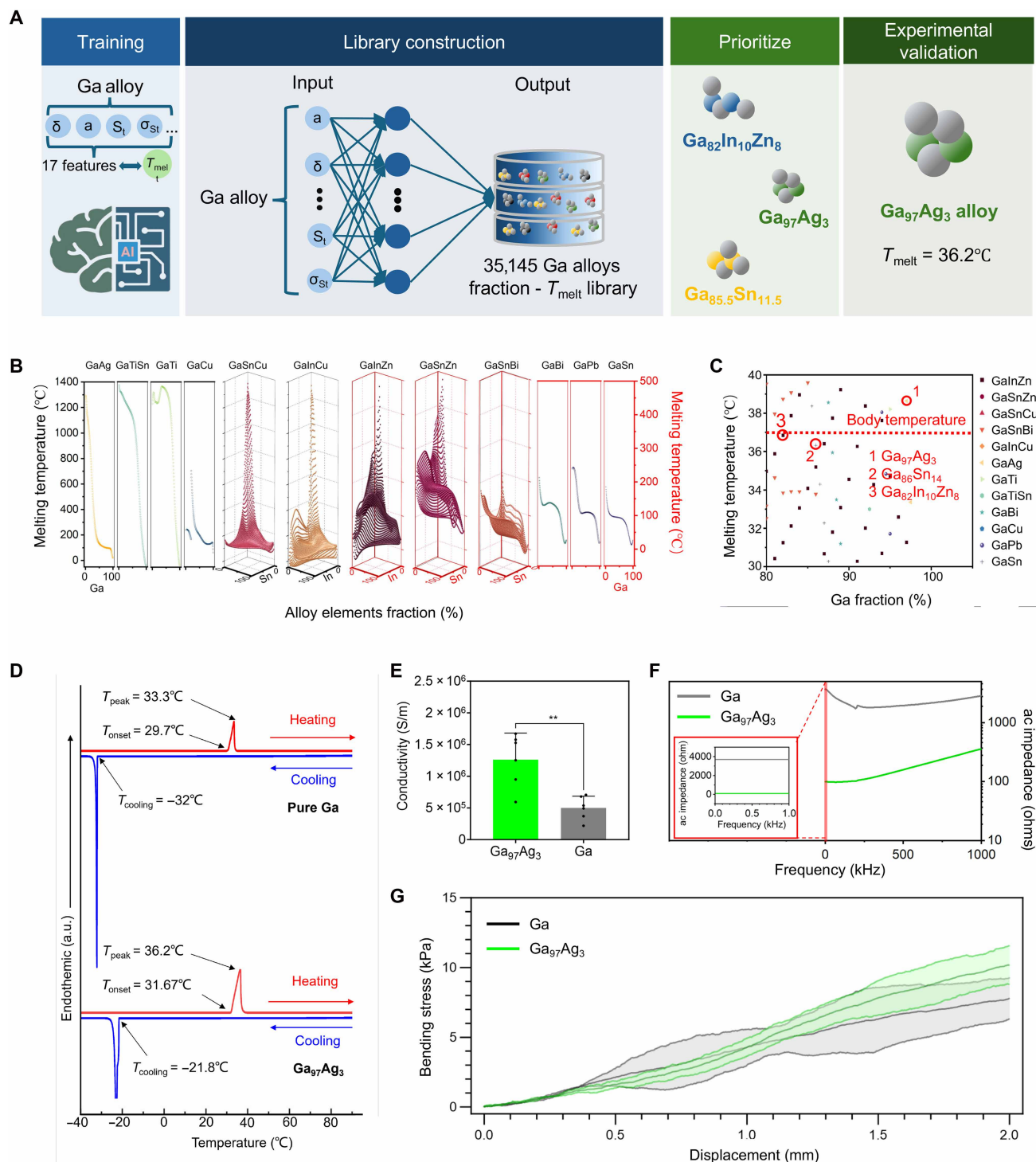


Fig. 2. AI-powered alloy design for optimized phase transition. (A) Workflow of the AI-powered LM-alloy design. (B) Library of all predicted LM-alloys and their melting temperatures. (C) Priority selection of Ga alloys that have a melting temperature close to body temperature. (D) Phase change characterization through DSC analysis. a.u., arbitrary units. (E) Conductivity measurements of $\text{Ga}_{97}\text{Ag}_3$ alloy and pure Ga ($n = 6$). (F) Impedance measurements of $\text{Ga}_{97}\text{Ag}_3$ alloy and Ga. (G) Bending stress of $\text{Ga}_{97}\text{Ag}_3$ alloy and Ga during a three-point bending test. ** $P < 0.01$, Student's t test.

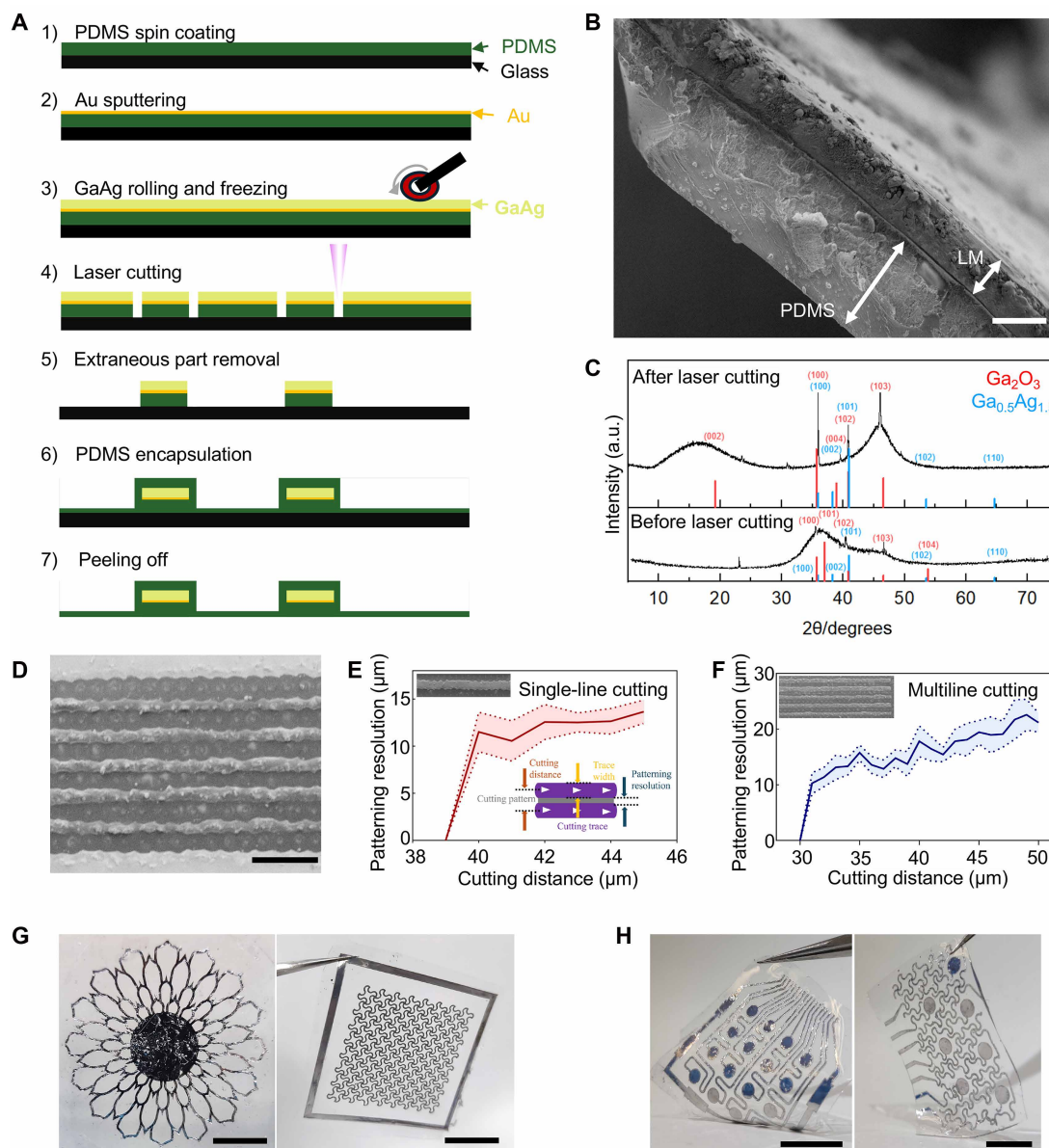


Fig. 3. Low-energy laser cutting for fabricating sFlex-Fold. (A) sFlex-Fold fabrication steps. (B) Side view taken by SEM of the edge cut by laser. Scale bar, 50 μm . (C) Comparison of the Ga_2O_3 by the XRD test. (D) SEM image showing a minimum linewidth of $\sim 11.5 \mu\text{m}$. Scale bar, 100 μm . Patterning resolution of (E) single-line cutting ($n = 16$) and (F) multiline cutting ($n = 16$). (G) Images of the patterning ability for complex patterns. Scale bars, 2 cm. (H) Fabricated sFlex-Fold electrode arrays. Scale bars, 1 cm.

PDMS onto a glass substrate and the subsequent sputtering of a 0.3-nm Au adhesion layer onto the PDMS. The substrate was then heated, and prewarmed GaAg was deposited onto the surface. The GaAg was then uniformly spread using silicone rollers. After freezing of the GaAg, the device underwent laser patterning. An ultraviolet (UV) laser was used to precisely cut both the electrode layout and the serpentine mechanical backbone of the device. After the laser cutting step, a Ga oxide film was observed to form at the cutting edge, which allows the GaAg to hold its shape (Fig. 3, B and C). The laser cutting step was observed to produce spherical particles at the cutting edge, showing composition matching the bulk Ga-rich alloy matrix of ~ 0.2 to 0.4% Ag (fig. S3, C and F).

Upon the removal of the excess GaAg, the fabrication of the sFlex-Fold device was completed through final encapsulation by another layer of thin film PDMS and its release from the substrate. PDMS was selected over other polymers, such as thermoplastic polyurethane (TPU) and styrene isoprene styrene (SIS), because it uniquely combines superior chronic biocompatibility, ultralow modulus for optimal tissue matching, and thermoset processing compatibility that preserves high-resolution liquid metal patterning without solvent or thermal distortion risks (29). The thickness was optimized to maximize the switchable flexibility coefficient, defined as the ratio of device modulus of the rigid phase or flexible phase while ensuring mechanical integrity and reliable encapsulation (fig. S6). The optimized

structure comprises $\sim 150\text{ }\mu\text{m}$ of PDMS and a $68\text{-}\mu\text{m}$ GaAg core (total thickness: $218\text{ }\mu\text{m}$), yielding effective moduli of 12 MPa (rigid) and 0.027 MPa (flexible), corresponding to a 440-fold mechanical contrast.

The resolution of the patterning approach is determined by the laser spot size used in the fabrication process. The smallest feature size is then determined by the step size between cutting lines while maintaining the line of LM between them. Our results indicate that the minimum linewidth achievable is $\sim 11.5\text{ }\mu\text{m}$ (Fig. 3, D to F), with a spot size of $\sim 30\text{ }\mu\text{m}$ and a step size of $40\text{ }\mu\text{m}$ between cuts. In this way, we can pattern a density of 20 lines/mm, which is adequate for making the circuits needed for epicortical recording applications. The utilization of UV laser cutting for patterning enables the creation of different patterns and electrode layouts tailored for various organs (Fig. 3, G and H). In designing the LM patterns, mechanical backbone and electrode routing were also optimized within the same layer. A serpentine mesh geometry was adopted to maintain continuous load paths for solid-phase rigidity, whereas localized routing and electrode features were defined within the remaining area to maximize channel density. The minimum achievable linewidth ($11.5\text{ }\mu\text{m}$) and routing density (20 lines/mm) set the fabrication-limited upper bound for electrode density, whereas backbone connectivity and metal fill fraction govern folding fidelity and shape retention in the rigid state.

In addition to the fully encapsulated sFlex-Fold, which was used for recording purposes in a noncontact format via capacitive coupling, a contact-based sFlex-Fold was also fabricated, capable of both recording and stimulation. A small platinum disk is used as the device-tissue interface, allowing for direct charge transfer with desired impedance values. Laser-cut platinum disks are embedded into the PDMS at the electrode point, with the legs of the platinum electrode pushing through the PDMS, contacting the GaAg interconnects (fig. S7). To prevent leakage, very small amounts of PDMS are used to secure the electrode legs in place.

Temperature-responsive switchable flexibility

The switchable flexibility capability of sFlex-Fold, guiding neural tissue conformation, is largely granted by an extreme change in the effective modulus of the device at the desired temperature ($\sim 37^\circ\text{C}$). Therefore, the mechanical properties of the device throughout a bi-directional transition between its rigid and flexible states was assessed (Fig. 4). In a demonstration of the rigid-to-flexible switching, a version of the sFlex-Fold that contains only the serpentine backbone was created. At room temperature, the device is rigid, easily holding its shape when lifted by tweezers. The device is then gently warmed to a temperature greater than 37°C to trigger the switching into its flexible state, resulting in a floppy, folded shape (Fig. 4A). This demonstrates a low effective modulus (1.66 MPa), desirable for the maintained in vivo conformation of implanted neural interfaces.

As previously described, the flexible-to-rigid switching of the device allows for shape memory. A 3D human brain model was created by reconstructing the magnetic resonance imaging (MRI) scanning data of a 70-year-old patient with Alzheimer's disease (AD) through high-resolution photoresin 3D printing (fig. S8). The device was shown to easily conform to the surface of the reconstructed 3D human brain model in its flexible state and to hold its shape once cooled down to -22°C and switched to its rigid state. The conservation of the exact curvature of the physiological brain structure was clearly observable once sFlex-Fold was lifted off the brain model, which could be directly translated to a real brain for 3D access to cortical

structures, where the device conforms to the region of both sulci and gyrus through its memorized shape and structure (Fig. 4B and movie S1).

To quantify the mechanical switching of the sFlex-Fold, both tensile stress tests and three-point bending tests were performed. These tests demonstrated a large contrast in the effective modulus of the device to which it attributes its performance at desired temperatures (22°C versus 37°C). In the tensile stress test the effective modulus dropped from 2.63 GPa to 1.66 MPa between its rigid and flexible states (Fig. 4, C and D). Similarly in the three-point bending test, the bending stress dropped from 0.276 MPa to 5.23 kPa at a 1.0-mm bending displacement (Fig. 4E). Conductance measurements during cyclic bending suggested that the sFlex-Fold showed no change in device conductance after 300 bending cycles. The conductivity of the device also remained unchanged between its two states.

The stability of the electrical interface of the sFlex-Fold was quantified through electrical impedance spectroscopy (EIS) during repeated mechanical deformation (Fig. 4, H to K). Devices were subjected to 10,000 cycles of four representative loading modes, including 90° torsion, 50% bending, 10% tensile strain, and 25% compression. In parallel, the impedance was monitored throughout as a measure of interconnection integrity. Across all modes, the electrodes maintained stable and functional impedance, with minimal variation under torsion and a slight decrease in impedance under bending, tensile, and compressive cycling, indicating stable solid-liquid contact between the LM and Pt without fatigue-driven degradation. A destructive tensile test further showed that electrical continuity was preserved up to $\sim 30\%$ strain, with failure occurring due to PDMS tearing at the pin anchoring site rather than loss of Pt-LM contact. Notably, the applied deformation amplitudes and cycle count substantially exceed physiological brain micromotion (30), providing a considerable mechanical safety margin for chronic implantation.

Ex vivo assessment of brain-tissue conformation in porcine brains

To examine the sFlex-Fold's ability to conform precisely to the sulcal and gyral structures of the cortex, a 3D study was conducted on an ex vivo model using porcine brains. The sFlex-Fold conformed to the cortex with gyri and sulci structures similar to human brains (Fig. 5A). For comparison, a traditional flexible electronic surface electrode array (hereafter termed CuPI-array; see details in Methods; fig. S9A) was also implanted at the same brain location. The CuPI-array [$12.5\text{ }\mu\text{m}$ of polyimide (PI) and $18\text{ }\mu\text{m}$ of Cu] was selected as a representative flexible bioelectronic baseline used in practical neural-interface implementations (31, 32). This choice enables comparison against conventional flexible electronics with gigapascal-scale polymer mechanics, thereby highlighting that sFlex-Fold's advantage derives from intrinsic modulus switching and tissue-matched softness rather than geometric thinning alone.

Once rigid-to-flexible switching was performed, the device's ability to conform to the brain's contours was assessed using a 3D reconstructed virtual model created by computed tomography (CT) scanning (Fig. 5, B and C). The virtual 3D model that was created enabled a quantitative assessment of the device's conformation to the brain surface. This assessment involved calculating both the gap distance between the device and the brain along different scanning axes (Fig. 5, D to K) and the maximum curvature angle that the device could achieve (Fig. 5, M and N). The gap distance quantifies the extent of the device's intimate physical contact with the brain surface and was derived by

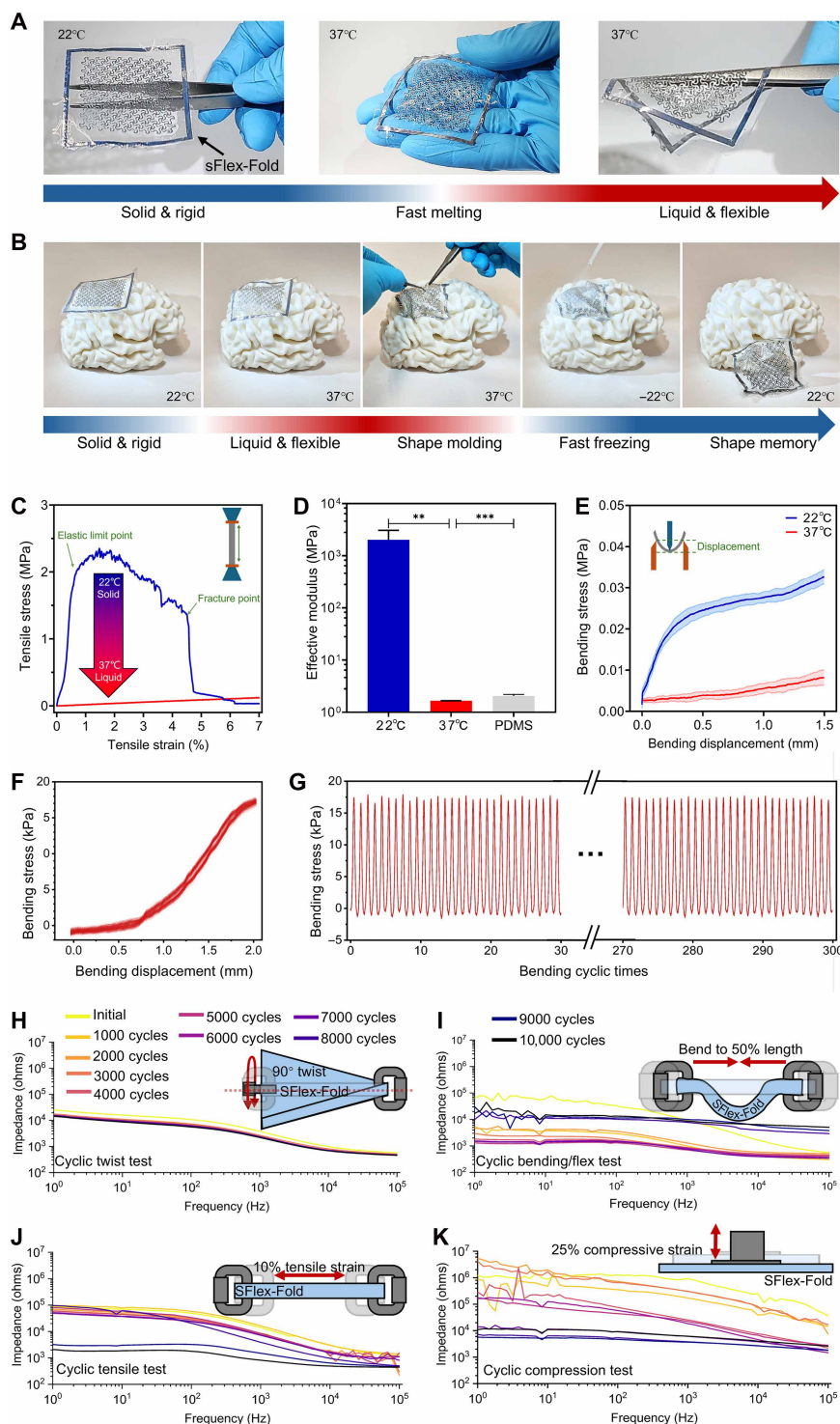


Fig. 4. Temperature-responsive switchable flexibility. (A) Rigid-to-flexible switching of sFlex-Fold through heating. (B) Rigid-to-flexible-to-rigid switching, demonstrating the shape memory of the cortex of a 70-year-old patient with AD with a reconstructed anatomical model. (C) Tensile stress test of $\text{Ga}_{97}\text{Ag}_3$ in PDMS at different temperatures. (D) Effective modulus of sFlex-Fold in both the rigid and flexible states, compared to pure PDMS of the same dimensions ($n = 3$). (E) Three-point bending test of sFlex-Fold ($n = 3$). (F) Bending stress versus displacement and (G) bending stress versus bending cycles. (H to K) Impedance spectra of the electrode under cyclic deformations (up to 10,000 cycles). The inset illustrates 90° torsional twist test (H), cyclic bending to 50% of original length (I), cyclic 10% tensile strain (J), and cyclic 25% compressive strain (K). ** $P < 0.01$; *** $P < 0.001$, Student's t test.

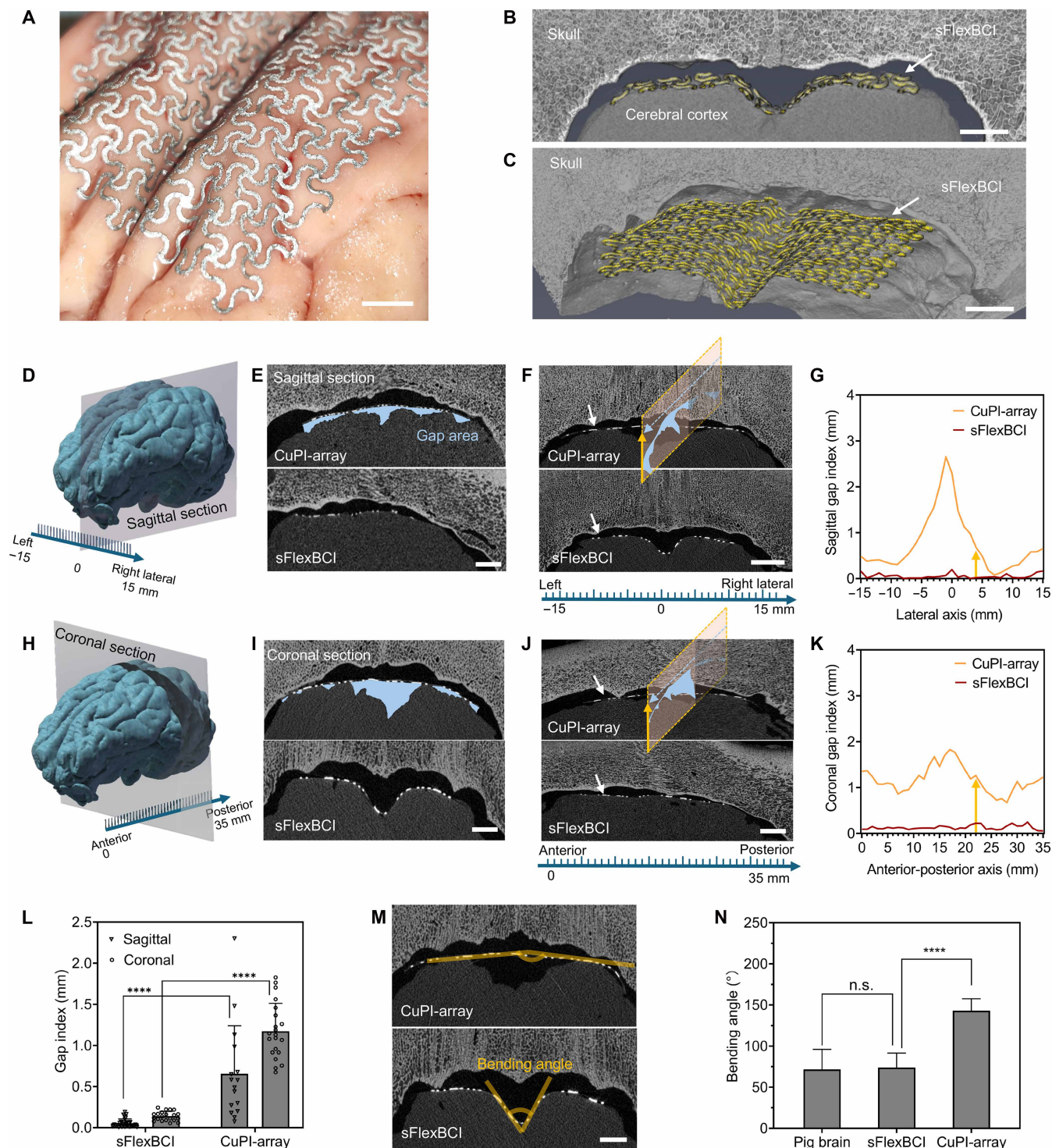


Fig. 5. Cortical conformation demonstration on an ex vivo porcine brain. The ability of the sFlex-Fold to match the curvature of the cortical surface has been assessed by quantifying the physical space between the device and brain tissue in 3D CT scans of sFlex-Fold on an ex vivo porcine brain and comparing its performance to a CuPI-array. (A) sFlex-Fold on the porcine cortex. (B and C) 3D reconstruction from the CT scan of sFlex-Fold on the porcine brain. (D) Illustration of sagittal sections along the lateral axis. (E) Sagittal sections of CuPI-array and sFlex-Fold with a space between the device and tissue in blue. (F) Illustration of the gap index calculation in sagittal sections along the lateral axis. (G) Calculated gap index of sagittal sections along the lateral axis. (H) Illustration of coronal sections along the A/P axis. (I) Coronal sections of CuPI-array and sFlex-Fold with a space between the device and tissue in blue. (J) Illustration of the gap index calculation in coronal sections along the A/P axis. (K) Calculated gap index of coronal sections along the A/P axis. (L) Average gap indices of the sFlex-Fold and CuPI-array along the two axes ($n = 31$ for sagittal sections and $n = 36$ for coronal sections). (M) Illustration of bending angle measurement. (N) Measured bending angles of the devices compared to porcine brains ($n = 36$ coronal sections). Scale bars, 5 mm [(A), (B), (C), (E), (F), (I), (J), and (M)]. *** $P < 0.001$; **** $P < 0.0001$; n.s., not significant, Student's t test.

quantifying the physical space between the device and the brain. Minimizing this gap distance is crucial for neural interfacing as a lack of intimate physical contact between the tissue surface and the electrodes prohibits high-quality neural signal recording and stimulation.

For each sagittal scan (Fig. 5, D and E), the total gap distance (tGAP) between the implant and the cortical surface was calculated through a summed measurement of all gap areas (Fig. 5, D to G). The tGAP was then normalized to the device length, which appeared in the scanning image to derive the gap index. Multiple sagittal scans were performed from left to right over a 31-mm span along the lateral axis, and the sagittal-tGAP was evaluated every 1 mm to give a quantitative index of the device conformation to the 3D brain structure (Fig. 5G). Similar quantification was also performed for coronal scans to evaluate the coronal-tGAP over a 35-mm span along the anterior-posterior (A/P) axis (Fig. 5, H to K). The sagittal gap index and the coronal gap index were both found to be greater than 1 mm for 48.5% of the scanning coverage for the CuPI-array (Fig. 5, G and K), suggesting the existence of large cavities under the implant coverage area, which would prohibit good electronic interfacing. In contrast, the gap index for the sFlex-Fold is only 0.055 mm for sagittal and 0.139 mm for coronal (Fig. 5L), which is a 10-fold improvement in conformal contact, indicating the capability of sFlex-Fold to perform high-quality neural recording and stimulation over a large area of the brain, especially for achieving immediate access to deep the grooves of sulci that are unattainable for the CuPI-array, which already assumes a silicone elastomer-based flexible construction. In addition, we also quantified the bending angle of device at the sulcal region (Fig. 5M), which confirmed that sFlex-Fold can match the curvature angles of the brain, whereas the CuPI-array could not (Fig. 5, M and N).

To evaluate potential Ag cytotoxicity and encapsulation integrity, Ag release from the GaAg-based devices was quantified by inductively coupled plasma mass spectrometry (ICP-MS) following 7-day incubation at 37°C in an agarose gel phantom mimicking the intracranial environment (fig. S10). No statistically significant increase in Ag concentration was detected in gel regions contacting the device compared to distant regions or phosphate-buffered saline (PBS) controls, with measured levels below reported physiological background concentrations.

In vivo characterization of recording and stimulation performance

Next, we assessed the ability for high-quality neural recording and stimulation provided by the conformal contact of sFlex-Fold to a rat brain in the central groove region (Fig. 6, A and B). Two types of sFlex-Fold, namely, noncontact (for recording only) and contact (for both recording and stimulation) (fig. S7), were created. These devices were used for recording electrocorticogram (ECoG) signals from the surface of the cortex, and their performance was compared to that of a commercially available ECoG-array (Pt circuit and electrodes embedded in PDMS; see details in Methods; fig. S9B). Specifically, the cingulate gyrus area 1 (Cg1) region of the rat brain lies within its central groove, and getting high-quality electrical recording or stimulation in this region of the brain is typically challenging for surface devices that are incapable of conformation to the cortical structure (Fig. 6D). Although the grooves present in rodent brains are much less pronounced than in large mammals, the central groove served as a sufficient analog for the demonstration of the key innovation of this work.

To validate signal recording ability (Fig. 6, C to F), neural signals were recorded from the groove sidewall of the Cg1 region using the sFlex-Fold as well as a commercially available flexible electrode device (ECoG-array) as a point of comparison for recording performance. The electrodes of the ECoG-array were fully encapsulated as the noncontact-sFlex-Fold was used. The ability of these two devices to capture the neural signals was assessed by comparing them to reference neural signal recording captured with a tungsten needle electrode in the opposite hemisphere (Fig. 6, C and D). The Cg1 region in rodents is known for exhibiting symmetric patterns of brain activity as the activity is synchronous across the two cortical hemispheres (33). This synchronicity was experimentally validated by comparing simultaneous recordings of the Cg1 region across both hemispheres, taken by tungsten needle electrodes (fig. S11). The similarity between the signal produced by the different electrodes (either sFlex-Fold, ECoG-array, or tungsten needle) and the reference signal was quantitatively assessed through a signal temporal correlation coefficient. The high coefficient achieved by sFlex-Fold indicates that it was able to successfully capture the neural signal from the Cg1 region in the central groove, whereas the ECoG-array was not (Fig. 6, C and E, 0.74 ± 0.07 versus 0.32 ± 0.08 , sFlex-Fold versus ECoG-array). The signal-to-noise ratio (SNR) of sFlex-Fold recordings was also comparable to that of the tungsten electrode and was significantly better than the ECoG-array signal (Fig. 6, D and F). This benefit is likely rooted in the fact that the sFlex-Fold interface was able to form intimate contact with the Cg1 region of the brain whereas the ECoG-array was unable to (Fig. 6C), and this benefit would be even more pronounced in large animal or human brains with more complex gyrification structures (Fig. 6D).

In addition to electrical recording of spontaneous brain activity, the improved conformation of sFlex-Fold was also assessed through its ability to specifically stimulate neural firing in the Cg1 groove region (Fig. 6, G and H). As the left and right Cg1 regions are synchronously linked for a high degree of interhemispheric communication (33–35), the success of neural stimulation of the right Cg1 groove should elicit robust neuronal activity in the left Cg1, which will be captured by an inserted tungsten needle electrode (Fig. 6, G and H). The electrophysiology link was confirmed by experiments using tungsten electrodes for both stimulation and recording at opposing Cg1 regions across the central groove, where a single stimulation in the right Cg1 could reliably induce a poststimulation local field potential (LFP) peak in the opposite hemisphere (fig. S12). For both the sFlex-Fold and ECoG-array device, the electrodes were exposed to contact the cortical tissues. The contact-sFlex-Fold was created through the addition of Pt contact pads to act as the conductive interface between the tissue and the LM circuitry (Fig. 6A and fig. S3). A single pulse was delivered via the device for stimulating the right Cg1 region (Fig. 6, G and I). Upon stimulation, a synchronized poststimulation field excitatory postsynaptic potential (EPSP) in the opposite Cg1 region was reliably detected by the tungsten recording (Fig. 6, G to K). The recorded LFP signal showed well-matched temporal association with the stimulation signal, as indicated by analyzing the pre- and poststimulation signal amplitude (Fig. 6K). When the stimulation pulse was delivered by the ECoG-array, no induced neural firing activity was specifically recorded in the opposite Cg1 groove (Fig. 6H), suggesting that the ECoG-array device was unable to form the electrical contact required with the sulci-like structure (Cg1 groove), whereas the sFlex-Fold was able to form such contact (Fig. 6, G and H). These observations demonstrate the ability of the

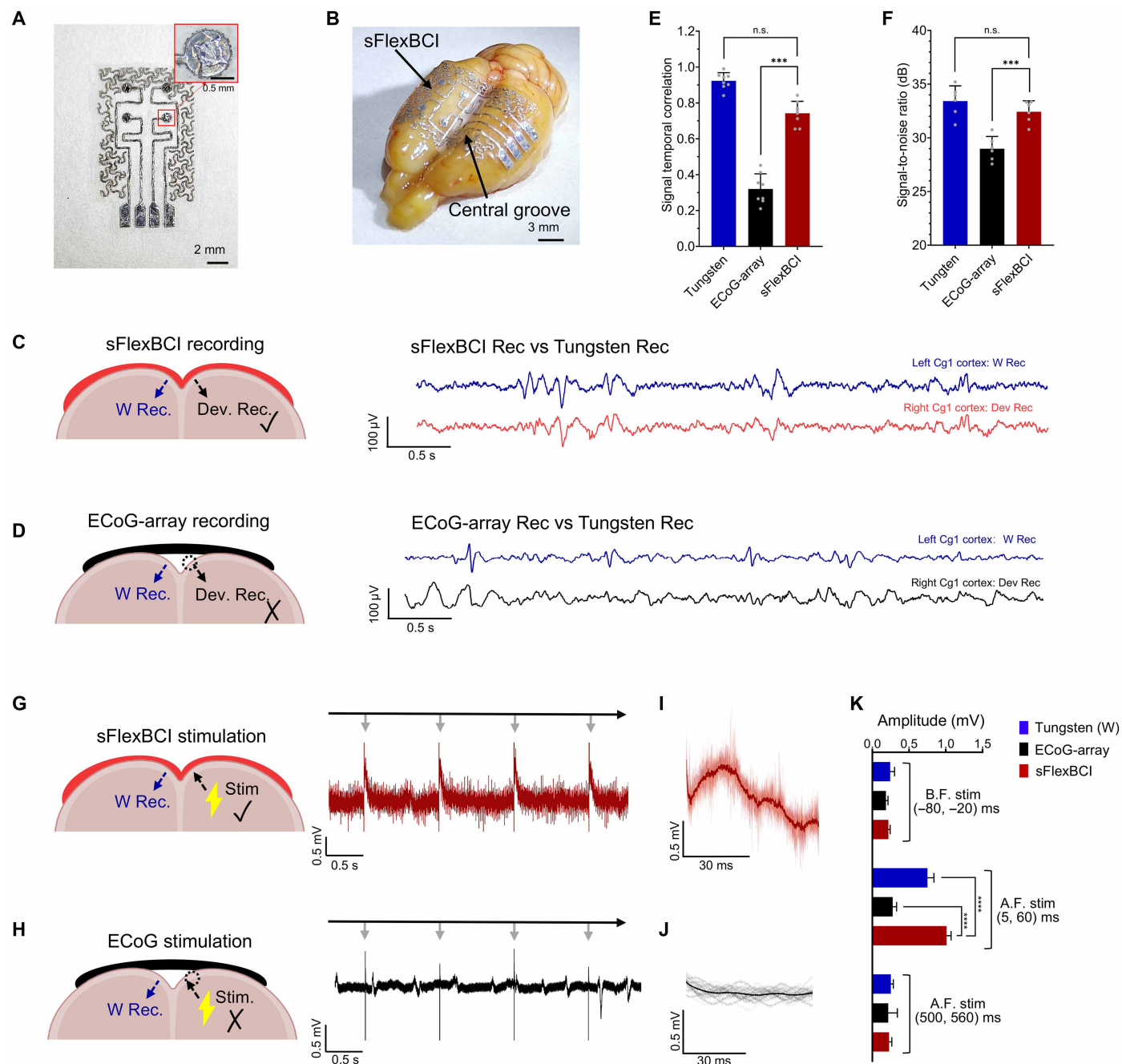


Fig. 6. In vivo recording and stimulation performance characterization. (A) Contact-sFlex-Fold with the Pt electrode. Scale bars, 0.5 mm (top) and 2 mm (bottom). (B) Ex vivo demonstration of sFlex-Fold conforming to and accessing the central groove. Scale bar, 1.5 mm. (C) Illustration of the sFlex-Fold recording experiment showing high signal correlation to the tungsten electrode in the opposite Cg1 region, indicating that the device under test (DUT) was able to successfully record signal from its respective Cg1 region. (D) Illustration of the ECoG-array recording experiment showing low signal correlation to the tungsten electrode in the opposite Cg1 region, indicating that the DUT was unable to successfully record signal from its respective Cg1 region. (E) Signal correlation analysis of tungsten, ECoG-array, and sFlex-Fold when compared to the signal recorded by a tungsten electrode in the opposite Cg1 region ($n = 8$). (F) SNR of signals recorded by the tungsten electrode, ECoG-array, and sFlex-Fold ($n = 6$). Illustration of cerebral cortex stimulation induced by (G) sFlex-Fold and (H) ECoG-array and the recording of evoked neural signal at the opposite Cg1 region. (I) Average waveform of the neural signal evoked by sFlex-Fold stimulation (0.4 V, 1 ms, single pulse, $n = 20$). (J) Average waveform of the neural signal evoked by ECoG-array stimulation (0.4 V, 1 ms, single pulse, $n = 20$). (K) Evoked signal amplitude of pre-, post-, and 0.5-s poststimulation by the tungsten electrode, ECoG-array, and sFlex-Fold ($n = 20$). *** $P < 0.001$; **** $P < 0.0001$; n.s., not significant, Student's t test.

perfect conformation afforded by the sFlex-Fold to allow for high-quality neural stimulation in hard-to-reach regions in brain grooves. Together, the ability of the sFlex-Fold to achieve high-quality recording and stimulation in the brain groove region of a rat model, combined with the quantitative assessment of the device's ability to match the curves of a brain in an ex vivo large animal model, would suggest the theoretical performance of the sFlex-Fold for in vivo studies in humans or large animals.

Advantages in long-term implantation

To evaluate the long-term biocompatibility of implanted sFlex-Fold, a 4-week subdural implantation experiment was performed. A sFlex-Fold and CuPI-array of the same thickness (68 μm and 98 μm ; fig. S9C) were implanted to the subarachnoid space covering the central groove region. In the following 24 hours after the implantation surgery, all the subjects implanted with sFlex-Fold survived, but most (75%) of the animals in the CuPI-array group died of acute intracranial hematoma (fig. S13), suggesting a significantly reduced surgical risk posed by the implantation of the sFlex-Fold. Comparing the surviving animals, a significantly better recovery was also observed for

sFlex-Fold group subjects, as indicated by pain level measurement [Rat Grimace Scale (RGS) score], weight changes, and movement and activity measurements (Fig. 7, A to E).

After 4 weeks, the sFlex-Fold still showed conformal contact with the cerebral cortex (Fig. 7, A and F), whereas the CuPI-array was left making little contact with the intended central groove region (Fig. 7, A and G). Structurally, the tissue under the sFlex-Fold shows much less sign of compressive distortion, in comparison to the CuPI-array subject where it was quite pronounced due to the motion between the skull and the cortex and the pressure from neural tissues under stress (Fig. 7, F to I). On the molecular level, the sFlex-Fold only induced slightly increased expression of the glial fibrillary acidic protein (GFAP) expression at the device-brain interface (Fig. 7, H and J), suggesting that the sFlex-Fold is less likely to induce any glial scar for better maintenance of the intended neural interface. The sFlex-Fold-implanted animals also showed almost no increased microglia activation, demonstrated through Iba1⁺ expression, suggesting no significant increase in inflammatory response (Fig. 7, I and K). The mechanical matching of sFlex-Fold (while in the flexible state) and conformal contact with the host brain tissue

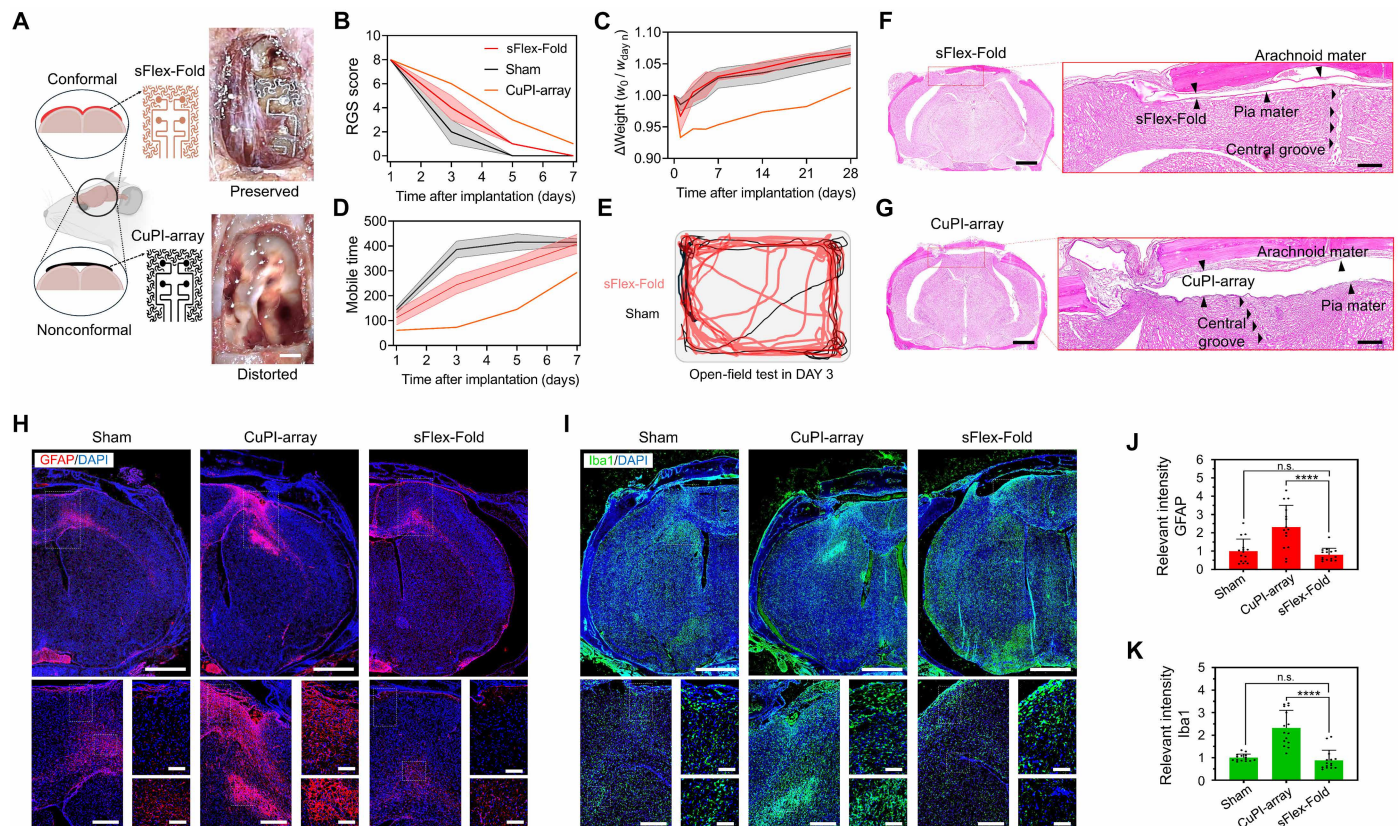


Fig. 7. Long-term subarachnoid implantation in rat brain. (A) Illustration of the long-term implantation. sFlex-Fold and CuPI-array are separately implanted; the histology and biocompatibility assessments are performed, respectively. Scale bar, 1.5 mm. (B) RGS score of the surviving subjects in the first week. (C) Weight change over 4 weeks postimplantation. (D) Animal's mobile time for every 10-min observation window at days 1 to 7 postimplantation. (E) Comparison of the mobile activity by the open-field test, showing movement trace from sFlex-Fold-implanted or CuPI-array-implanted rat at day 3 postimplantation. Hematoxylin and eosin (H&E) staining of (F) sFlex-Fold-implanted and (G) CuPI-array-implanted rat brain after 28 days. Scale bars, 1 mm (left) and 200 μm (right). (H) GFAP immunofluorescence staining for sham surgery, CuPI-array-implanted, and sFlex-Fold-implanted rat brain samples. (I) Iba1 immunofluorescence staining for sham surgery, CuPI-array-implanted, and sFlex-Fold-implanted rat brain samples. For (H) and (I), the top panel shows an overall view of a large tissue section staining (scale bars, 2 mm); the bottom-left panel shows the enlarged view of the outlined region in the top panel (scale bars, 200 μm); the bottom-right panels show the further enlarged view of the outlined regions in the bottom-left panel (scale bars, 200 μm). (J) Statistical analysis for the relevant intensity of GFAP ($n = 15$) and (K) Iba1 ($n = 15$). **** $P < 0.0001$; n.s., not significant, Student's t test.

play an important role for reduced glial scar formation and neuroinflammation with regard to a fully implanted neural interfacing system. Four-week in vivo implantation showed minimal GFAP up-regulation and no significant Iba1 activation, supporting effective encapsulation and the absence of detectable toxic leaching over the tested period.

DISCUSSION

Here, we present the design and development of a neural interface device, sFlex-Fold, capable of structurally and mechanically matching the cortex, providing stable electrical coupling to large cortical regions including those regions hidden within sulcal folds. Typically, devices that have aimed at similar levels of deep tissue interfacing have to rely on penetrating implants (36–39), and simultaneous large coverage of the protruding gyri of the cortex is then impossible (12–18). sFlex-Fold acts as the device that is capable of nondestructive 3D neural access with both large area and nonpenetrating deep tissue coverage. This is achieved by the development of a bioelectronic system with switchable flexibility, awarding sFlex-Fold the unique advantages inherited from both rigid and flexible modalities. sFlex-Fold can be morphed into complex, folded, 3D shapes, enabling in vivo implantation into the deep cortical sulci while maintaining large coverage of curved brain surfaces. When switched to its flexible state postimplantation, it further allows for the electronics to maintain electrical and mechanical tissue conformation for intimate neural interfacing and minimal in vivo damage.

Similar to recent presentations of flexible electronic arrays (12–18, 36), this work aims to demonstrate a bioelectronic device that structurally and mechanically matches the cortex but takes an alternative approach. The pioneering work by Viventi *et al.* (40) demonstrated the value of ultrathin geometries for improving conformability to the cortical surface and enabling large-area, high-density mapping. Unlike the popular solution of ultrathin sheet electrode arrays for conformation (36, 41), the monolithic sFlex-Fold is not implanted in its flexible state, nor does it rely on an auxiliary implantation shuttle (42). The device can be directly applied onto the cortical surface, and its 3D and folded structure ensures that insertion, orientation, and conformation are inherently achieved. Although prior shape-changing or morphing devices have primarily sought to increase generic cortical coverage through deployment or actuation (17, 18), sFlex-Fold applies its phase transition to preserve a predefined, anatomy-specific 3D geometry matched to the targeted cortical region. The AI-tuned liquid metal core enables a rigid state for self-supporting sulcal access without auxiliary shuttles or deployment mechanisms, followed by an automatic transition to a tissue-matched compliant state upon reaching body temperature. In doing so, it resolves the long-standing contradiction between devices stiff enough for sulcal reach and those soft enough for chronic biocompatible coupling.

The use of Ga-based LMs has been recently explored for the purpose of flexible bioelectronics (43–47). However, researchers are yet to fully capitalize on the unique phase change properties of LMs for their use in electronics as of yet, although there have been a few reports describing the concept of using the LM phase change for transient mechanical properties (48–51), for example, the development of intravenous needles that irreversibly soften upon insertion (51). Needlelike electrodes have also been developed for use in electrophysiology studies (48, 49). These studies show great potential in creating LM-based transient electronic systems; however, the application of

this idea to create 3D-conformed electrode arrays remains previously unknown. As part of material innovation in this work, an AI-powered alloy design system was used to optimize the flexibility switching of the sFlex-Fold by shifting the melting point of the LM-alloy core toward body temperature (37°C). The alloying study presented here outlines a pathway for seeking new LM-alloys tailor-made for specific applications, even beyond biomedical implants. It should be noted that the alloying model established in this study is preliminarily used as a general screen of potential candidates rather than a true predictor of perfect alloy formulations. Future work could expand the capabilities of the model as an alloy predictor by more comprehensive construction of the training samples of different alloy types (e.g., eutectic, intermetallic, etc.).

A fundamental engineering trade-off in sFlex-Fold arises from the dual role of the LM layer: Increasing electrode density requires finer pitch and reduced spacing, whereas preserving accurate 3D shape memory in the solid phase demands a sufficiently connected structural backbone and adequate areal metal fraction. In this work, these constraints were balanced by using a mechanically continuous serpentine mesh that ensures global rigidity in the solid state while enabling high-resolution circuit routing within fabrication limits. Future studies can be expanded to systematically optimize mesh geometry, trace width, pitch, and fill factor, potentially guided by computational modeling (52), to scale channel count while preserving folding fidelity and mechanical robustness for anatomy-specific designs. For example, an improved implementation of the device can incorporate a transition from the discrete Pt electrode insert to batch-scalable approaches, such as thin-film Pt/PtIr metallization, screen-printed Pt deposition, or lamination-based contact formation. In addition, although the current ~11.5- μm patterning resolution is sufficient for large-area, surface-conformal applications demonstrated here, refined intracortical adaptations could leverage higher-resolution microfabrication strategies (e.g., reduced spot-size laser systems or photolithographic LM patterning) to enable finer electrode geometries without altering the core bistable design concept.

The chronic compatibility is especially important when considering the prospect of treating complex brain disorders with advanced neural interfaces (3, 25, 26). Once implanted, sFlex-Fold of reduced modulus further allows the adaptive capability to maintain intimate electrical contact with large-coverage cortical tissues even as brain structure changes. For example, the progress of AD is notoriously featured by cortical thinning, reduced gyrification, and increased sulcal width (26). It is therefore paramount to develop devices that can deform with the changing neural landscape and maintain electrical contact. Beyond disease management, the interfacing provided by sFlex-Fold provides the ability to decipher the signaling and connectome of human brain through direct measurement of signals from both gyri and sulci cortex, which could potentially inspire the next generation of neuromorphic computing (53). As the gyri and sulci allow for dense neural networks in the human brain, neuromorphic chips can use 3D designs or layered architectures to maximize the density of artificial neurons and synapses, without increasing the footprint.

MATERIALS AND METHODS

Materials

The materials used for fabricating the sFlex-Fold are mostly acquired from Sigma-Aldrich. The materials from other vendors are specified accordingly.

ANN model training

Our ANN models were trained using MATLAB's Deep Learning Toolbox (MATLAB R2023b). The network follows a feedforward architecture, consisting of an input layer, a hidden layer with 15 neurons, and an output layer. The activation function used in the hidden layer is the sigmoid function, whereas the output layer uses a linear activation to predict the melting temperature.

The dataset comprises 173 samples, with 70% used for training, 15% for validation, and 15% for testing. As input, 17 data descriptors are normalized using the formula: $x_i = \frac{x_i^0 - x_{i\min}^0}{x_{i\max}^0 - x_{i\min}^0}$, ($i = 1, 2, \dots, 17$), where x_i and x_i^0 represent the normalized and initial values of the i th descriptor, respectively. $x_{i\max}^0$ and $x_{i\min}^0$ represent the maximum and minimum values of the i th descriptor, respectively. The output is the melting temperature.

To evaluate the model, we used metrics such as RMSE and R^2 . A fivefold cross-validation was also performed to ensure the robustness of the results. All training and testing procedures were conducted on a system equipped with an Intel Core i7-12700H processor, 16-GB RAM, and an NVIDIA RTX 3060 GPU.

Synthesis of Ga alloys

The Ga alloys were prepared through vacuum induction melting under a high-purity argon atmosphere (WK-II Melt Spinner, Physics). Different ratios of high-purity Ga (99.90 wt %) and other high-purity alloy components (over 99.9 wt % purity) were placed in a quartz tube for melting. To minimize contamination, the sealed and mounted quartz tube was flushed with high-purity argon (purity: 99.999%) before melting. The samples were heated until melted and held for an additional 3 min to ensure chemical homogeneity. After heating, they underwent air cooling to room temperature.

DSC for Ga alloys

The DSC measurements were carried out on a thermal analysis System (DSC 3, METTLER TOLEDO) under a high-purity argon atmosphere (purity: 99.999%) with a flow rate of 50 ml/min. The temperature range was set from -100° to 100°C with a heating rate of 2 K/min and a cooling rate of 4 K/min. Samples (weighing ~ 50 mg) were placed into alumina crucibles during measurements. The instrument was calibrated using indium and zinc standards to ensure temperature and enthalpy accuracy. Baseline correction was performed before measurements. Data analysis was performed using the STARe software (version 17.00) to determine thermal events, with onset and peak temperatures determined using tangent intersection methods.

Microstructural and elemental characterization for the Ga₉₇Ag₃ alloy

To evaluate the phase homogeneity of the bulk alloy and investigate the composition of features at the laser-processed edges, microstructural characterization was performed using a field-emission scanning electron microscope (FEI Quanta 450 FEG) equipped with an EDAX TSL EDX system. The imaging and elemental analysis were conducted at an accelerating voltage of 20 kV. Samples were mounted on aluminum stubs using conductive carbon tape to ensure electrical grounding. Two specific regions were targeted for analysis: (i) the as-cast, unpatterned Ga₉₇Ag₃ alloy surface to determine the intrinsic phase distribution and (ii) the laser-cut edges to identify the origin of the observed particulate features. EDX elemental mapping

was used to visualize the spatial dispersion of Ga and Ag. Furthermore, quantitative point-mode EDX analysis was performed to calculate the atomic percentage of the constituents within the matrix, the needlelike/particle-like intermetallic phases, and the spherical resolidified droplets, thereby distinguishing between intrinsic alloy phases and processing by-products.

Four-point conductivity measurement

Thin films of GaAg and pure Ga were prepared by spreading the materials onto 7 cm-by-7 cm glass substrates. High-purity GaAg alloy and Ga (99.99%, Sigma-Aldrich) were used. Specifically, both the metal materials and glass were preheated to 90°C with a plate heater. Approximately 50 μl of the liquid-phase metal was dropped onto the glass surface and then was rolled out with a silicone roller. The resulting samples were frozen in a -30°C refrigerator.

The prepared samples were then mounted on a multifunction digital four-probe tester (ST2258C, Jingge Electronics) for conductivity measurements. The probe distance was set to 1 mm, and all measurements were conducted under an applied voltage of 12 V at room temperature ($\sim 25^\circ\text{C}$). The conductivity (σ) was calculated using the standard four-point probe equation: $\sigma = \frac{I}{V} \cdot \frac{\pi}{\ln(2)} \cdot \frac{1}{t}$, where t represents the film thickness.

ac impedance measurement

The ac impedance of GaAg and pure Ga samples was measured using the current-voltage (I - V) method. The samples were fabricated into bars with dimensions of 0.3 mm by 3 mm by 40 mm (± 0.1 mm). The surfaces of the samples were polished and cleaned using ethanol and deionized water to ensure good electrical contact. An ac voltage of 5 V [root mean square (RMS)], with a frequency range from 1 Hz to 1 MHz, was supplied by a waveform generator (33500B, Keysight). And the actual voltage on the samples (V_{ac}) was recorded using an oscilloscope (DSO-X 3012A, Agilent Technologies), whereas the current through the samples (I_{ac}) was measured using a precision multimeter (8846A, Fluke). Then, the ac impedance (Z) was calculated by the formula: $Z = \frac{V_{\text{ac}}}{I_{\text{ac}}}$, where V_{ac} and I_{ac} represent the RMS values of the measured voltage and current, respectively. The measurements were performed at room temperature.

Cyclic mechanical test with electrical characterization

To evaluate the long-term electrical reliability of the sFlex-Fold device under dynamic mechanical deformation, a custom-built electromechanical testing platform was established. The setup comprised a single-axis reciprocating linear actuator (model: J110-JXD096-053-035-T200-01, Just Motion Control, China) integrated with a steel optical breadboard featuring an M6 tapped hole array. Custom-designed multi-degree-of-freedom crab-like clamps were anchored to the breadboard to secure the device, allowing for flexible reconfiguration between different deformation modes.

The device was subjected to four distinct mechanical loading protocols to simulate physiological strains and handling stresses: (i) torsional twisting with a rotation angle of 90° ; (ii) bending/flexing, where the device was compressed to 50% of its original length; (iii) tensile stretching at 10% strain; and (iv) compression at 25% strain. For each mode, the device underwent a total of 10,000 mechanical cycles at a frequency of 2 Hz.

To monitor the evolution of the electrode-interconnect impedance, EIS measurements were performed intermittently every 1000 cycles. The EIS data were recorded using a CH1760e Electrochemical

Workstation (CH Instruments Inc.) in a PBS solution at room temperature. The impedance spectra were swept over a frequency range of 1 Hz to 1 MHz. In addition, a destructive tensile test was conducted by stretching the device until electrical failure to determine the ultimate mechanical limit.

Three-point bending test and cyclic bending test

The mechanical difference of GaAg and pure Ga was compared using the three-point bending test. The rigid phase samples were created with dimensions of 0.3 mm by 3 mm by 40 mm (± 0.1 mm).

The mechanical properties of sFlex-Fold containing were evaluated using a three-point bending test and a cyclic bending test. The samples were created, with same dimensions, and encapsulated in a thin film of PDMS (thickness: 0.1 mm) to provide structural support and prevent oxidation. The three-point bending test of sFlex-Fold was produced separately during its rigid and soft phase.

The three-point bending test was conducted using an Instron 5942 Micro Tester equipped with a 50-N load cell (accuracy: 0.01 N). The samples were loaded on an Instron 5942 Micro Tester with a three-point bending setup and the support span of the two supporting pins. The bending was applied at a constant speed of 10 mm/min until a maximum displacement of 4 mm was reached. During the test, the force-displacement curve was recorded.

For the cyclic bending test, the maximum displacement was set to 2 mm, and the loading speed was increased to 60 mm/min. Each sample was subjected to 300 cycles, with the force-displacement response monitored.

Tensile test for phase changing characterization

A tensile test was conducted to assess the mechanical changes of the sFlex-Fold during its liquid and solid phases. Specifically, fully encapsulated sFlex-Fold measuring 40 mm by 3 mm by 0.3 mm was mounted on the clamp-on strain sensors of a material testing machine (5942 Micro Tester, Instron, USA). The device was then stretched in either the liquid or solid phase at an arm movement speed of 10 mm/min until breakage occurred. The effective modulus (E) was calculated by dividing the tensile stress (σ) by the tensile strain (ϵ).

XRD test

To verify the fabrication of GaAg and Ga₂O₃ film formation, x-ray diffraction (XRD) test was conducted to determine the crystalline structure of the GaAg alloy and assess the postcutting sample for oxide film generation. Samples were prepared with a layer of GaAg film (300 μ m in thickness) on a glass substrate with a diameter of 3 cm. A subsample then underwent laser cutting by a UV laser with low-energy parameters (7 A, 1500 mm/s, and 10 passes) to create cutting traces. Then, the precutting and postcutting samples were prepared for the XRD test. XRD raw data were acquired on a Bruker D2 phaser XE-T x-ray diffractometer system by using Cu K α radiation ($\lambda = 1.5406$ Å), over a 2θ range of [5°, 75°] with a step size of 0.01° and a scan rate of 0.5°/min. Data processing was performed using the MDI Jade 9.0 software. The phases were identified by comparing the diffraction patterns with reference data from the Powder Diffraction File (PDF4+ 2009).

SEM imaging for patterning characterization

The cutting edge and patterning resolution of the sFlex-Fold were evaluated by SEM. To optimize the patterning resolution, 24-mm-diameter circular samples were prepared as per steps 1 to 3 of Fig. 3A.

A series of rectangular patterns, each 5 mm in length and with widths ranging from 35 to 45 μ m, was created to assess the patterning resolution. This was to assess how close the cutting traces could be while still being individually resolved. In addition, a series of raster-shaped patterns, each 5 mm in length and with single rectangular units ranging in width from 25 to 55 μ m, was fabricated to evaluate the multiple line patterning resolution, again assessing how close the cutting traces could be while still being individually resolvable. The laser cutting parameters were set to a cutting current of 5 A, a cutting speed of 1500 mm/s, and 100 cutting passes.

The samples were then loaded into an environmental SEM (e-SEM) system (FEI Quanta 250, Spain). SEM imaging was performed at a magnification of 500x, with an accelerating voltage of 2.00 kV and a chamber pressure of 7.85×10^{-4} Pa. The length and width of the patterned lines were measured using the ImageJ software, and the data were analyzed using the Origin 2021 software ($N = 3$).

To assess the cutting edge, samples were collected from the extra-neous areas of the sFlex-Fold. These samples were subjected to the same SEM imaging procedure as described above to evaluate the quality and precision of the cutting edges.

Laser cutting parameter optimization

To determine optimal laser cutting parameters for the highest resolution, a test sample was prepared as described in steps 1 to 3 of Fig. 3A. Lines of 5 mm in length were cut into the GaAg with different cutting parameters, imaged by a stereomicroscope, and analyzed by the ImageJ software. Specifically, the cutting current was tested from 5 to 10 A, and the cutting speed was tested from 500 to 2000 mm/s.

After cutting, the sample was imaged using a stereomicroscope to capture a detailed view of the cutting traces. The images were then analyzed using the ImageJ software to measure and evaluate the resolution of the cutting traces by assessing how well the laser was able to cut all the way through the GaAg.

Comparison devices

Different comparison devices were used throughout this study to accurately assess the performance of sFlex-Fold in key areas. Namely, for the cortical conformation demonstration on the ex vivo porcine brain, a CuPI-array (18- μ m copper layer on a 12.5- μ m PI substrate) was fabricated to act as a comparison flexible electronic device. Both the sFlex-Fold and the Cu-PI-array were cut into 50 mm-by-50 mm serpentine mesh shapes (Fig. 4A and fig. S5A). The CuPI-array was also used as a comparison device to sFlex-Fold for the long-term implantation study. Here, both the sFlex-Fold and CuPI-array were patterned into a 9.4 mm-by-10.5 mm design containing both a serpentine backbone and an electrode layout suitable for a rat brain (Fig. 7A).

To assess both the stimulation and recording performance of the sFlex-Fold a commercially available ECoG-array (eight-channel KD-FECOG, KedouBC) with exposed Pt electrodes (50 μ m in diameter) was used as a comparison device. The same pattern of the sFlex-Fold was used for recording and stimulation (Fig. 6A). The commercial ECoG-array is encapsulated by PDMS, and during the recording experiment, the Pt electrodes were covered by extra PDMS to mimic the noncontact version of sFlex-Fold. For the stimulation comparison, the contact-sFlex-Fold was used (fig. S3) and the electrodes of the ECoG-array were left exposed.

Fabrication of the sFlex-Fold

sFlex-Fold was fabricated via a process of laser cutting GaAg and encapsulating it within thin PDMS. The process began with the preparation of a PDMS film. A 12:1 ratio of PDMS base to curing agent was mixed and then spin-coated onto a 10 mm-by-10 mm glass substrate using a spin coater (KW-4T, IMECAS). The spin-coating process involved two steps: an initial spin at 500 rpm for 30 s and then ramped up to spin at 1300 rpm for 10 s. The coated PDMS film was then cured on a plate heater at 80°C for 40 min.

Subsequently, a 0.3-nm gold layer was deposited onto the cured PDMS film using a gold sputtering system (Q150TS Dual Target Sputtering System, QUORUM). GaAg was then spread over the gold layer using a roller on a 50°C hot plate. The thickness of the liquid metal film was controlled by placing layers of tape along the edges of the glass substrate. The GaAg coated substrate was then frozen in a −30°C refrigerator to solidify the metal layer.

The patterned sFlex-Fold was created using a UV laser cutting machine (HDZ-WAF600 laser engraving machine, HAN'S LASER). The laser cutting parameters were set to a cutting current of 8 A, a cutting speed of 1000 mm/min, and 200 passes. After laser cutting, the extraneous areas were removed, and another layer of PDMS was spin coated on top of the patterned layer using the same parameters. This second PDMS layer was also cured on an 80°C plate heater for 40 min.

Last, the sFlex-Fold was carefully peeled off from the glass substrate while maintaining a temperature of 50°C on the plate heater to facilitate the release process. For the contact-based sFlex-Fold, a circular Pt electrode with four pins was used. The pins were designed to penetrate through the PDMS layer, contacting and adhering to the electrode area of the LM layer, ensuring reliable electrical connections.

Fabrication of the CuPI-array

PDMS with a 12:1 mixed ratio was spin coated (500 rpm for 30 s and 1300 rpm for 10 s) onto a 10 mm-by-10 mm glass substrate. Then, a 13 mm-by-13 mm CuPI film (18 μm of Cu and 12.5 μm of PI) was attached on the PDMS with its PI side and rolling by a roller (to extrude air between the film and the PDMS). Then, it was placed on an 80°C heater for 40 min to cross-link the PDMS. After being patterned by UV laser (10 A, 1000 mm/min, and 50 passes), the extraneous part was removed, and another layer of PDMS was fabricated on the Cu side by spin coating and heating following the same procedure as the first layer.

Comparison of thermal stability and flexibility switching kinetics

To evaluate the specific contribution of the alloying strategy to device handleability, control devices were fabricated using pure Ga following identical encapsulation protocols and dimensional specifications (metal and substrate thickness) as the Ga₉₇Ag₃ devices. Thermal stability was assessed by placing the rigidified devices (pre-set to a memorized shape) on a skin surface (~32°C) to simulate intraoperative handling conditions. The capacity of the device to retain its 3D geometry versus softening was monitored visually. To quantify the phase transition speed, a flexibility switching test was conducted in a water bath maintained at 37°C, simulating the intracranial thermal environment. The flexibility switching time was defined as the duration required for the device to transition from a rigid, self-supporting state to a fully flexible state under gravitational bending upon immersion.

Measurements were performed on four independent samples ($n = 4$) for both the pure Ga and Ga₉₇Ag₃ groups.

Geometric optimization and mechanical characterization

To maximize the mechanical contrast between the rigid and flexible states while ensuring encapsulation reliability, a systematic geometric optimization of the sFlex-Fold was performed. The thickness of the PDMS encapsulation layer, which primarily governs the flexible-phase response, was tuned by adjusting the secondary spin-coating speed (10-s duration following an initial 30-s spread at 500 rpm). Samples with varying PDMS thicknesses were fabricated by adjusting the speed to 800, 1000, 1200, and 1400 rpm. Independently, the GaAg alloy feature thickness was varied from 34 to 85 μm to determine its impact on the rigid-phase modulus. Mechanical characterization was conducted using an Instron 5942 Micro Tester at a constant strain rate of 30 mm/min. Young's moduli were calculated from the linear regions of the stress-strain curves, corresponding to strain ranges of 0 to 0.05% for the rigid phase and 0 to 4% for the flexible phase. The switchable flexibility coefficient was derived as the ratio of the rigid-phase modulus to the flexible-phase modulus ($E_{\text{rigid}}/E_{\text{soft}}$). The final geometry (150 μm of PDMS and 68 μm of alloy) was selected to achieve a balance between a low soft-phase modulus and robust, leak-free sealing.

Ex vivo cortical conformation CT measurement on the porcine brain

The conformability of the sFlex-Fold was evaluated using CT measurements to analyze the gap index between the sFlex-Fold and the brain surface. Initially, a porcine head was dissected, with the skin and muscle removed, leaving the skull and brain intact. The top portion of the skull was then removed, and the sFlex-Fold, along with a CuPI-array as a control, was implanted before replacing the skull fragment. The ex vivo implantation models were scanned using a Micro CT (uCT100, SCANCO, Switzerland). The scanning parameters were set to a voltage of 70 kV, a current of 200 μA, with a resolution of 34.2 μm and an exposure time of 300 ms. The μCT Ray V4.2 imaging processing software was used to output the original TIF image files and construct the 3D reconstructed model.

Assessment of Ag ion leakage and encapsulation integrity

To evaluate the long-term encapsulation integrity of the GaAg-based devices and assess potential Ag ion leakage, an ex vivo diffusion study was performed using an agarose gel phantom. The sFlex-Fold devices, encapsulated in PDMS, were embedded within the gel phantom to simulate the intracranial physiological environment and incubated at 37°C for 168 hours. Following the incubation period, samples were collected from the immediate device-gel interface, distant gel regions (denoted as "Empty"), and a PBS control group. The concentration of Ag in each sample was precisely quantified by ICP-MS (Avio 220 Max ICP-OES) to detect any trace leakage of Ag species under these simulated physiological conditions.

Animals

All animal experiments were conducted in accordance with protocols approved by the Animal Ethical Committee of City University of Hong Kong (CityU) and the Department of Health of the Government of HKSAR. Male and female Sprague-Dawley rats, aged 8 to 12 weeks, were acquired from the Laboratory Animal Services Centre of CUHK and maintained in the Laboratory Animal Research Unit (LARU) of CityU, following standard guidelines.

Neurosurgery procedures for cortex recording, stimulation, and long-term implantation

The rats were anesthetized by intraperitoneal injection of pentobarbital (50 mg/kg, Ceva Santé Animale). An additional intraperitoneal injection of atropine (3.25 mg/kg) was administered to reduce bronchial secretion that could cause suffocation. Once the rats were mounted on a stereotaxic apparatus (Narishige), an incision was made on the scalp to expose the skull, followed by disinfection with 2% iodine and 75% ethanol. A square craniotomy window (5 mm by 10 mm) was created in the cortical region [window center at 0-mm A/P and 0-mm medial-lateral (M/L)] to allow the placement of the sFlex-Fold. Two small access holes (1.3 mm in diameter, 6-mm A/P and 1-mm M/L, and −12-mm A/P and 0-mm M/L) were drilled into the skull to insert screw electrodes, serving as ground and reference electrodes of the signal recording system for both the cortical stimulation and recording experiments. Throughout the experiment, a heating pad was placed beneath the animal to maintain a stable temperature of 37°C.

In the recording and stimulation experiments, a tungsten needle electrode was used to measure the LFP from the left cg1 region in the central groove (1.2-mm A/P and 0.5-mm M/L), whereas the electrode regions of both the sFlex-Fold and ECoG-array were applied to the right cg1 region (1.2-mm A/P and 0.5-mm M/L) to characterize their ability to capture the synchronous signal.

In experiments assessing the long-term biocompatibility of the sFlex-Fold, the rats underwent the same surgical procedures as described above. This experiment included the implantation of either sFlex-Fold or CuPI-array with patterns shown in Fig. 7A. After implantation, the skull fragment was reset, and the brain aponeurosis and scalp were sutured separately. The surgery site was disinfected following the same procedure as described above. During the 4-week recovery period, survival rate, RGS score, weight changes, and open-field movement trace data were collected. After the recovery period, the rats were euthanized for histological analysis.

Neural signal recording

For electrophysiology recording, the sFlex-Fold was connected to a 36-pin flexible flat cable (FFC) socket soldered onto a customized printed circuit board (PCB), facilitating connection to a preamplifier (TDT). Specifically, the sFlex-Fold implant and tungsten needle electrodes on both sides were directly connected to a mini-DB26 headstage (preamplifier). The headstage was then connected to the PZ5 neurodigitizer for signal amplification and digitization. The neural signals were recorded using the Synapse software (TDT). Data from all channels were transmitted to an RZ base station for further processing via a single fiber optic connection. During recording, the sampling frequency was set at 24,414 Hz to ensure high-resolution signal acquisition, with a 50-Hz notch filter applied to remove powerline noise.

A customized MATLAB program was developed to process the recorded signals, calculate the signal correlation coefficient, and determine the SNR separately.

Cerebral cortex stimulation in rats

To facilitate electrophysiology recording, the tungsten needle electrode was stereotactically positioned in the left cg1 region of the rat brain, as defined by Paxinos and Watson's rat brain atlas as previously described.

To assess their stimulation performance, a tungsten needle electrode, sFlex-Fold, and ECoG-array were connected to the cathode of

a waveform generator (33500B, Keysight) through a custom interface, whereas the anode was connected to a scalp electrode near the stimulation site and external ground. The sync channel of the waveform generator was connected to the RZ2 base station to provide time markers for synchronization. The electrical impedance of each device was measured at 1000 Hz and was found to be 0.26 ± 0.1 k Ω for the sFlex-Fold, 0.24 ± 0.05 k Ω for the ECoG-array, and 1.2 ± 0.2 k Ω for the tungsten needle electrode. On the basis of their contact area and impedance, single-pulse stimulation was configured as follows: 1-ms single pulse electrical stimulus with a −400 mV amplitude (sFlex-Fold and ECoG-array) and −2 V amplitude (tungsten electrode). Each trial consisted of a single manual trigger followed by 20 pulses at a frequency of 1 Hz.

The neural signals were recorded via a tungsten needle electrode through the method described for neural signal recording above. The stimulation signal was divided into 1-s windows (−0.2 to 0.8 s) based on the stimulation marker channel. The script performed waveform superposition and calculated the average response waveform and maximum amplitude. Data visualization and statistical analysis were performed using custom MATLAB functions and the Origin 2021 software.

Histology and immunofluorescence assay

Long-term implantation samples were collected 4 weeks postimplantation. The rats were deeply anesthetized with 2% pentobarbital (2.5 ml/kg, intraperitoneally) before perfusion with PBS and 4% paraformaldehyde (PFA). The samples containing the skull, device (either sFlex-Fold or CuPI-array as in Fig. 7A), and brain were carefully harvested to maintain structural integrity and fixed in 4% PFA overnight at 4°C. Following fixation, the samples were dehydrated in graded sucrose solutions (10, 20, and 30% in PBS) until the samples settled at the bottom of the container. The samples were then decalcified in 10% EDTA decalcification solution at a controlled temperature of 25° to 30°C for 27 days, with the solution replaced every 3 days. Decalcification was confirmed by testing tissue softness. After decalcification, the samples were embedded in OCT compound and sectioned into coronal slices (10 μ m in thickness) using a cryosectioning instrument (Cryostar NX70, Thermo Fisher Scientific). The slices were stored at −80°C until further use.

For immunofluorescence staining, all procedures followed a standard protocol. The sample slices were blocked in tris-buffered saline (TBS) supplemented with 0.5% Triton X-100 and 3% bovine serum albumin (BSA) for 2 hours. The samples were then incubated with primary antibodies (1:250) diluted in blocking solution at 4°C overnight. The primary antibodies used included anti-Iba1 (ab48004, Abcam) and anti-GFAP (ab7260, Abcam). Each primary antibody was applied to different tissue sections. Following primary antibody incubation, the sample slices were gently rinsed with PBS (3 $\times$ 5 min) and incubated with secondary antibodies (1:500) diluted in TBS at room temperature for 2 hours. The secondary antibodies used included anti-Rabbit IgG H&L (Alexa Fluor 488, Abcam). After PBS washing, the samples were stained with 4',6-diamidino-2-phenylindole (DAPI; 1:500 dilution in TBS, 20 min at room temperature) to label cell nuclei. Last, the stained sections were mounted with antifade mounting medium for imaging.

Confocal microscopy (Leica SP8) was used to acquire fluorescence images with the following parameters: excitation wavelengths of 488 nm for Alexa Fluor 488 and 405 nm for DAPI. The fluorescence intensity of each marker was normalized to the mean intensity

of the sham control group. The ImageJ software was used for background subtraction, thresholding, and intensity quantification. The relative fluorescence intensity was represented as a fold change compared to the control.

Statistical analysis

All experimental data in this study were subjected to statistical analysis and are presented as means $\pm$ SD. Before statistical analysis, all data were tested for normality using the Shapiro-Wilk test to determine whether parametric or nonparametric tests were appropriate. For comparisons between two groups, an independent two-tailed Student's *t* test was used if the data followed a normal distribution. If the data did not meet the assumption of normality, a nonparametric test such as the Mann-Whitney *U* test or Kruskal-Wallis test was used. Statistical significance was defined as $P < 0.05$. Statistical significance is noted in all figures as follows: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and **** $P < 0.0001$. All statistical analyses and data visualization were performed using the Origin 2021 software. Data fitting, graphing (e.g., bar plots and scatterplots with error bars), and statistical tests were conducted within the same software environment.

Supplementary Materials

The PDF file includes:

Figs. S1 to S13

Legends for movies S1 and S2

Table S1

References

Other Supplementary Material for this manuscript includes the following:

Movies S1 and S2

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Electronics with switchable flexibility for 3D conforming neural interfaces

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